A Mobile-Based Screening and Refinement System to Identify the Risk of Dyscalculia and Dysgraphia Learning Disabilities in Primary School Students

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***Abstract*—Learning Disability is a condition that has a direct effect on the brain and there is no cure or any identified medical treatments. Most of these cases remain undiagnosed due to the lack of awareness from their parents and teachers in underdeveloped countries like Sri Lanka. Mobile application-based solution 'Nana Shilpa' was developed for the screening and intervention processes for the specific Learning Disabilities which are Verbal and Lexical Dyscalculia, Operational and Practognostic Dyscalculia, Letter Level Dysgraphia and Numeric Dysgraphia. Deep Learning with Machine Learning techniques is used in the screening process to provide a better solution. To detect the written letters/numbers, trained Convolutional Neural Networks (CNN) achieved the accuracy of 92%, 99%, 99% for Verbal and Lexical Dyscalculia, Letter Level Dysgraphia and Number Dysgraphia respectively. The Machine Learning algorithms used for screening processes are Support Vector Machine (SVM) and Random Forest (RF). In the machine learning models, it is achieved the accuracy of 98%, 97% for Operational and Practognostic Dyscalculia and Number Dysgraphia respectively. In Sri Lanka, this has been recognized as an acceptable solution for screening and intervention via a mobile-based application for above mentioned four variants of learning disability conditions which are developed based on the gaming environment.**

Keywords— Learning Disabilities, Dyscalculia, Dysgraphia, Convolutional Neural Network, Machine Learning, Mobile application

I. INTRODUCTION

Learning disabilities (LDs) itself is not a very cordial term for under developing countries like Sri Lanka. Due to the lack of awareness and knowledge about learning disabilities, many students face discomfort in their learning process. In the majority of situations, students with learning disabilities are eliminated from further educational opportunities and neglected from achieving the learning outcomes. Even though Sri Lanka has achieved the 99.11% in the primary school enrollment rate as reported by [1], we have yet to acquire a long journey in establishing a quality inclusive education where students with disabilities can be identified and treated in the early stages refer to [2]. The accurate rates of students who are undergoing learning disabilities cannot be found since many cases of the disability occurrences are not verified or identified medically.

Due to these learning disabilities, students may develop aggressive behavior, school violence acts or victims of school bullying also. Refer to [3] learning disabilities had increased the risk of getting bullied by their peers and also social, emotional and behavioral damage which affects their growth and development of life.

These challenges can be addressed through a conducive learning environment. Through this research we have approached, an assistive Learning Environment that will improve the learning experience of LD students in their academic life. Learning handicaps stem from defects in one or more understanding, reasoning, recalling or learning-related processes which include language processing; phonological processing; space visual processing; the speed of processing; memory and attention and executive function. Concerning [4], the education metrics, Sri Lanka almost achieved the Millennium Development Goals (MDGs). Given 98 percent of elementary school enrolment (World Bank, 2013), however, many disabled children tend to be excluded and still remain as the subjects of high attrition levels. For example, 23.5% of school children (5-14 years of age) with handicaps are excluded from education [4].

Dysgraphia is a neurological disability condition where the students show unclear and irregular handwriting, difficulty in writing in provided space and slow writing compared to their peers. Refer to [4] the cause of this disability yet to discover but it has been stated that it is the result of the damage of the parietal lobe of the brain.

According to [5],[6],[8],[9], Dyscalculia is a term that can be described as a specific Learning Disability affecting the ability of an individual to understand numbers and discover more in mathematics. Dyscalculia is sometimes called number blindness. Much of the work into the development of mathematical skills has centered upon counting or arithmetic, but other complex skills also require these skills at early levels including: [5] knowing the word number (one, two, ...),

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numerals (1, …12, … 20, …) and their interrelationships. As stated by [7], Dyscalculia has distinction inside itself. Lexical dyscalculia is the variant that causes trouble in reading mathematical symbols. Verbal Dyscalculia is the variants that trouble in name mathematical quantities, numbers, and symbols. Operational Dyscalculia condition causes troubles in performing mathematical operations and calculus and Practognostic Dyscalculia causes troubles in enumerating, manipulating, and comparing real objects and pictures. These 4 variants of Dyscalculia have been used in the application in the screening and intervention process of Dyscalculia.

This research project was conducted to develop a mobile application for screening and intervention in the risk of dysgraphia and dyscalculia learning disability in primary school students in Sri Lanka. The fundamental objective of delivering this mobile application is to cater the accessible screening and intervention method for these risk of learning disabilities and elevate the basic skills of mathematical operations in calculations, reading, and Sinhala and number writing in primary school students.

II. LITERATURE REVIEW

Previous researchers have been conducted in recent years to develop an assistive tool for identifying dyscalculia in children. The "Dyscalculia Screener" is a program designed to evaluate the mathematical capacity of children 6-14 years old [10],[11]. It checks the children's wicker zones by giving various questions such as arithmetic operations in mathematics and comparison. But the major limitation was it supports only screening. In another web-based solution, students had been checked in calculations, numeric concepts, and problem-solving areas. But this is not a very attractive computer-based screener for children to use [8].

"DyscalculiUm" is a web-based solution for a screening test that evaluates children with dyscalculia [12]. But it assesses students who take part in higher education only. For the screening section for this diagnose, it was used six key factors to identify. Test items were made to prove that. Refer to [13], they have developed five different cognitive tasks to screen dyscalculia patterns associated with two systems, and screening test is done by using a tablet to mobilize the tool to enable the data collection.

The “Dynamo Maths” is computer software that can be used for intervention and providing a remedy for dyscalculic children who are in the 6 to 9 age range [14]. The three-stage program includes lesson plans on multi-sensory resources, interactive online modules, and worksheets for practice. The research, which was done in Thailand, the main purpose was to develop the assistive technology tool called “Calculating Aid Tools: KidKanit” to help dyscalculia students in mathematical areas [15].

The study [17], proposed a web-based learning tool as a design to improve the learning environment of different technologies of Dyscalculia students. It was only provided simple and very attractive game-based activities for the pre-school students. But there was no dyscalculia identification process. Refer to [16], “Dyscalc” is a free online screening tool for students over the age of 14. Furthermore, it is also a web-based application. But nowadays the most effective way of learning things using mobile [19]. A quick (2 minute) paper-and-pencil tool was developed to evaluate the ability of children to compare symbolic and non-symbolic numerical magnitudes to resolve these outstanding problems and evaluated the degree to which individual differences in achievement are explained by success on this test [13].

A study was done for the diagnosis of dyscalculia using three classifiers rule-PART, J48, and SVM to predict the different classes of Dyscalculia. The result was got based on 8 features. Three models were created using collected data and founded SVM had better resulted in the classification of various subclasses with a higher accuracy value is 94% compared to the other two classifiers [20]. “Calculic Kids” is a mobile application-based solution to learn math concepts efficiently which is conducted using four subsets of games. But the application does not contain a proper way for progress tracking and remedy activities [21].

Research work on developing an intelligent system to diagnose and suggest remedial measures for learning disorders in children is based on the data input of child’s behaviour provided by the parents, system diagnoses the disorders and suggest the remedial measure. According to [22], The main aspect of this system is, it collects child behaviour through a questionary by their parents to diagnose disorders and provide remedies.

The role of the mobile application developed in research [23] is to improve the hard life of persons with learning disabilities. The application supports individuals with learning disabilities to practice arithmetic and language disabilities to improve their brain efficiency. The contents of the computer-aided application designed consist of counting skills, place value concepts, and additional subjects which are related to mathematics lesson’s learning outcomes of primary school children as mentioned in [24].

In [25], [26], researches were conducted on identifying Dysgraphia situation and provide practice writing tasks. In [25] Time Warping algorithm and the expert system [26] have been implemented using Forward Chaining Method (FCM). Identified defects can be considered as achieved less accuracy, lack of attractive features, and providing only screening part.

AR-based Dysgraphia Assistance Writing Environment (AR-DAWE) modal was developed according to [27], which addressed spelling mistake issues of Dysgraphia students. Researches which are on identifying handwriting issues of the Dysgraphia students on [28] and [29], practical questionnaire and standardized Handwriting Proficiency Screening Questionnaire (HPSQ) are used for the screening process, respectively. These researches help only for spelling issues.

According to [30], a game that is based on the client-server architecture in which student can collect fruits by drawing different shapes circles, line. The key design principles which mainly focused were improved security for the user, interactive and informative guidance, high availability. An innovative approach system is proposed in which user can view the performance of the individual student or overall view of the class of the students which are allocated to them. The performance status of the student is evaluated through the initial conducted activities and depending on the condition of the student, the teacher may choose the most appropriate tool model for the student [31].

The research [32], system contains 3 main components like knowledge base, interface engine, user interface. In this system identifying the Dysgraphia condition is done through the decision tree where the input is analyzed by the rules

which reside inside the knowledge acquisition or knowledge base of the system.

To overcome these limitations and produce more accurate and reliable results, 'Nana Shilpa' delivers a unique mobile-based solution for screening and intervention for these two learning disability conditions.

III. METHODOLOGY

The "Nana Shilpa" mobile application system conducts the screening and refinement process of risk of dyscalculia and dysgraphia disability condition using 4 major risk identification and intervention components which are Verbal Dyscalculia, Operational and Practognostic Dyscalculia, Letter level Dysgraphia and Numeric Dysgraphia.

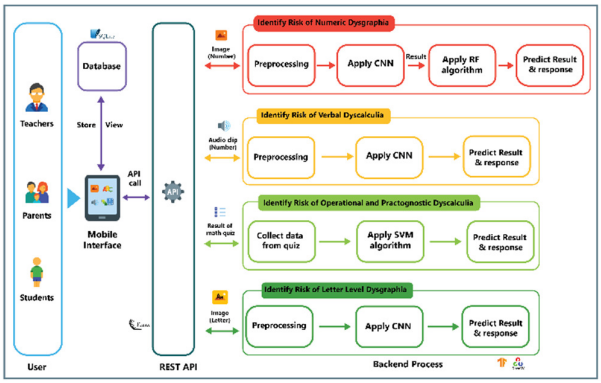


Fig. 1. System Overview

A. *Identify the risk of letter level dysgraphia*

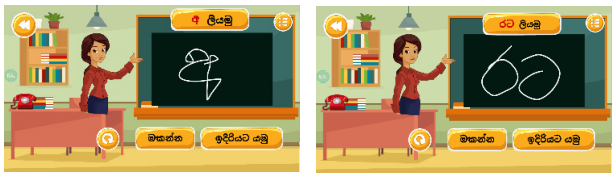


Fig. 2. Letter level dysgraphia screening

The Letter level Dysgraphia screening process has been conducted using four letters and four words in two stages shown in Fig. 2. These four words and four letters were selected on the consultation of primary school teachers and teachers guide of Sri Lanka local Sinhala syllabus. The selection of letters and words were ensured that it enfolds all the sections of the alphabet. To make the screening writing process more accurate and accomplished, students are asked to write letters and words at a time, and they are facilitated with clear the canvas and undo the movements features at the screening activity user interface.

The CNN model was built using the letter and word data images which were collected from non-dysgraphic and dysgraphic children from the age range 6 to 10. The model consists of 4 conv2D layers, 2 MaxPooling2D layers, a flatten layer and 2 dense layers. The conv2D layers contain 64 filters with a kernel size of 5x5 and 3x3 strides. The maxpooling2D layer contains a pool size of 2 and with 2 strides and a dense layer with 500 units. With an additional utilization process, a trained CNN model is used for the accurate screening of the risk of letter level dysgraphia condition.

In the screening process which involves the letter and word identification through the CNN model, initially, the image is being preprocessed by converting it to greyscale, contrast is improved by applying Histogram Equalization and undergo with the binarization process. In the segmentation process, letters and words are set apart and the loss function is being used as the Multi-class Cross-Entropy of the CNN model.

To screen the risk, letter level accuracy probability of written letters and words, total correct count of the letters and words and total erase count are being fed to the SVM model and utilized output results are used to determine the risk of having letter level dysgraphia. These results which include the date which the test has been performed, risk probability of having Letter Level Dysgraphia are being stored as the report in the database for the further reference of the parents and teachers.

Upon the result of having the risk of letter level dysgraphia, extensive intervention activities are provided to the student. The intervention activities are provided in two stages which are for intervention for letters and words. Letters and words are given to the students with the proceeding path which involves animations show from the start point to the endpoint of the path. The total process is conducted with background voice guidance and a mission-based story. The process is designed to obtain a more user-friendly and interoceptive learning process in which students can experience more positive outcomes.

B. *Identify the risk of operational & practognostic dyscalculia*



(a) Enumerating quiz (b) Operation: addition

Fig. 3. Operational and Practognostic Dyscalculia screening

In Operational and Practognostic Dyscalculia screening, an interactive mathematical quiz covering various areas was carried out to identify the children who suffer from dyscalculia or non- dyscalculia. Initially, enumerating (counting) quiz were given as shown in Fig. 3.a, and in the next stages comparison of numbers and objects quiz as well as the mathematical operation-based quiz covering the addition and subtraction carried respectively refer to Fig. 3.b.

In training the model, 225 quiz results were gathered from both dyscalculic and non-dyscalculic school children for each stage of the screening test, within the age group 6 to 8. The trained model was created using the scikit-learn machine learning library in python. A screening test for dyscalculia was developed which uses a question bank to carry out the test. In each stage of the screening test, students get various questions that are stored in the SQLite database on a random basis.

Here, performs classification by finding the hyper-plane that differentiates the two classes by using a SVM supervised learning algorithm. SVM predicts whether the students or children are having dyscalculia or not by using six

parameters. Along with the SVM Radial Basis Function (RBF) kernel was used to train the most optimal model to get a more accurate result since the data set is non-linear.

To screening the risk of having dyscalculia, the score of the mathematical operation-based quiz covering the addition and subtraction and average time to complete the mathematical operation-based quiz, the score of the enumerating (counting) quiz and an average time to answer the enumerating quiz, and finally sore of the comparison of numbers and objects quiz and an average time to complete the comparison of numbers and objects quiz will be taken as input to the model. These six features fed to the trained SVM model. The SVM model will then identify the child as regular or dyscalculia affected child. Further, screening results can be seen by selecting an attempt of the screening report of the student.

If the student is at risk of dyscalculia set of comprehensive activities with a multisensory learning environment will be given. It was further divided into four categories to practice mathematical addition operation, mathematical subtraction operation, enumerating, and comparison of numbers and objects as shown in Fig. 4.a. In mathematical addition operation there are four stages and eventually in each stage students will get the different concept of addition. In the first stage, students will be given a simple exercise to add two numbers and choose the corresponding answer represented in the blue Graphical card as shown in Fig. 4.b. After finishing the practice quiz students can review the quizzes that student has done. Likewise, all the categories have questions to improve the student's skills. After consultation with medical professionals, the remedy was planned to enhance the observation and skills of Operational and Practognostic Dyscalculia children.



(a) Remedy category menu (b) Remedy activity quiz

Fig. 4. Operational and Practognostic Dyscalculia remedy activities

C. *Identify the risk of numeric dysgraphia*



(a) Numeric Dysgraphia identifying test (b) Numeric Dysgraphia Screening Results Interface

Fig. 5. Numeric Dysgraphia Component Interfaces

Implemented 'Nana Shilpa' application provides both screening and intervention processes including more other features. The drawable canvas is provided by the application for the screening process. The application requires to write digits from 0 to 9 each twice as shown in Fig. 5.a to reach for more accuracy level. While writing digits, the correctness, erase count, time taken to complete the task as well as boundary touch count is taken per question to the Number Dysgraphia prediction. The result is provided as a probability of having Number Dysgraphia.

A CNN model is trained to check the accuracy of the written digit. Using the Modified National Institute of Standards and Technology (MNIST) number dataset imported from Keras, the model is trained. 19 layers and 20 epochs were used to achieve high accuracy from the CNN model. After trained the CNN model, it is downloaded as a TensorFlow lite file which is more convenient to use in the android environment. After writing a digit, a written digit image is converted to a Bitmap file and go through the added TensorFlow lite file. It returns the identified digit from the provided digit. If it is identified as correct, that quiz is correct. There are all 20 questions including from 0 to 9 each twice. After completion of the screening quiz, the results are shown which are total correctness, total erase count, total time is taken and total boundary touch count. Here, total correctness is based on how many questions are correct from 20 questions. Quiz results are shown as Fig. 5.b. A ML model is trained to predict the result using the results provided by the screening task. Based on the trained data in the ML model, it returns the result as a probability of whether a student has a risk for Numeric Dysgraphia or not. To catch the progress of a student, a parent can view the report which includes done quiz results.

If the implemented application predicts that there is a probability of having Numeric Dysgraphia, the system will redirect to the intervention interface. The intervention interface is provided in a gaming environment where the student can write digits from 0 to 9 with guidance. For each number, it provides graphical images which represents the written number, after done the writing. The process of intervention is implemented to obtain improved motor skills and observation in children who are undergoing the dysgraphic condition with the help of educational and medical specialists.

D. *Identify the risk of verbal and Lexical dyscalculia*



(a) Lexical Dyscalculia identifying test (b) Verbal Dyscalculia identifying test

Fig. 6. Verbal & Lexical Dyscalculia identifying test

In the Lexical Dyscalculia screening test, the application will provide background voice guidance and display a random number from a predefined number set of 12 which were chosen from 0 to 1000, considering 3 categories with the ones, tens, hundreds as Fig. 6.a. User has to press the microphone button and pronounce the displayed number. The user needs to perform this test five times and at the end, the application will show the risk percentage of Lexical Dyscalculia. The fact consider here is whether the user identifies the displayed number and pronounces it correctly.

A CNN used with five layers to identify Lexical Dyscalculia risk. It consists of three convolutional layers with batch normalizer, Maxpool2D layers, and "ReLu" activation function in each. The output flattens into 1D using a dense layer that consists of 64 neurons and a dropout layer and it feeds into another dense layer with 12 neurons and a softmax classifier. The final output goes through "Adam optimizer" and sparse categorical cross-entropy is used as the loss function for better results.

CNN model trained using pronunciations of 12 chosen numbers with 3 parts as train, validation, and test. 0.1 of the data set (10% of the whole) taken as test data, 0.1 from remaining data (9% of the whole) as validation. The rest has taken as the train split. Before feeding into the CNN model, a set of preprocessing steps followed. MFCCs of the audio file extracted as the feature for audio classification which is the best and most commonly used. Take snapshots at different segments of audio and used 13 MFCCs using the Librosa library.

In the Verbal Dyscalculia identifying test, with background voice guidance and recorded audio of the pronunciation of the number is played and user needs to select the correct answer played in the audio as shown in Fig. 6.b. The user is required to perform the activity five times and the risk of Verbal Dyscalculia percentage will be calculated using the data of the activity against the voice record played in the activity. The screening process is conducted to identify whether the user identifies the correct number according to voice pronunciation. The data related to each activity will be added to the database after performing and displaying the identifying test result.

Remedy practices will provide an example activity of how to pronounce & select the correct number. Users can choose from three examples which are given from three categories with the ones, tens, hundreds each. The application will train the user on how to pronounce the number and give time to pronounce the number and also select the pronounced number from different answers as shown in Fig. 7.

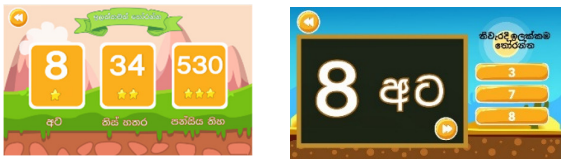


Fig. 7. Verbal & Lexical Dyscalculia remedy activity

IV. Result and Discussions

'Nana Shilpa' has been developed for screening and intervention of Dysgraphia and Dyscalculia in Sri Lankan primary school students using the students around different regions across Sri Lanka. The mobile application is implemented using android, SQLite database, Flask and Tensor flow as the middleware technology and the backend of the application is python. The research has been used machine learning concepts and algorithms, image processing, voice recognition techniques and frequency comparison in the implementation.

A. *Letter level dysgraphia identification*

More than 1000 letters were collected for each word and letters from primary school students and preprocessed using image processing techniques. CNN has been chosen since it is designed to map 2D or multi-dimensional image data to a one-dimensional output variable and it delivers high accuracy for the image classification and recognition process. In the model training process, the CNN model had obtained over 98% of accuracy for the test input data and Fig. 8.a shows the training and validation accuracy against 8 epochs of the model.

B. *Operational and practognostic dyscalculia identification*

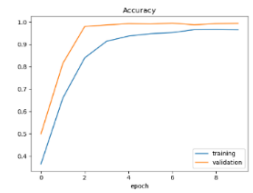
The collected data were fed into the SVM model as training data. The SVM classification algorithm was trained using the above mentioned six features to predict whether the children have dyscalculia or not. Nearly 98% accuracy was achieved identification of Operational and Practognostic dyscalculia as shown in Fig. 8.d. Among the 45 testing instances, 44 instances correctly classified which is 97.78% accuracy of the testing, and only 1 instance was predicted as incorrectly.

C. *Numeric dysgraphia identification*

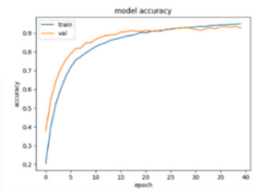
For image recognition and processing, CNN is used specifically to process pixel data. CNN was trained using the MNIST number dataset which imported from Keras achieving 99% accuracy. It is used over 60,000 training image datasets and 10,000 test dataset each from 0 to 9. In the ML model training, nearly 400 data were trained using different algorithms to find out what is best. Among RF, SVM, K-Nearest Neighbors and logistic regression which showed 98%, 90%, 89%, 88% accuracy respectively, RF was chosen for Numeric Dysgraphia Prediction.

D. *Verbal dyscalculia identification*

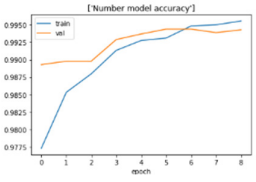
Initially, selected 12 numbers for the screening process by considering ones, tens, hundreds. Data needed for the CNN training process gathered from primary school students aged between 7 – 8. The audio data set consists of 400 .wav files from each. CNN model accrued 91% of test accuracy as shown in Fig. 8.b and 0.32 test error in the training process with 40 epochs.



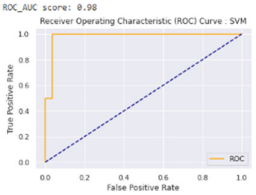
(a) CNN accuracy against 8 epochs for Letter Level Dysgraphia



(b) CNN accuracy against 40 epochs for Verbal Dyscalculia



(c) CNN accuracy against 8 epochs for Numeric Dysgraphia



(d) ROC curve for Operational and Practognostic Dyscalculia

Fig. 8. Accuracy evidence

VI. Conclusion

In the Sri Lankan context, very limited researchers have conducted on mentioned learning disability areas. To cater

for the requirement, 'Nana Shilpa' a mobile-based solution was implemented in a native Sinhala language for the screening and intervention processes for Dyscalculia and Dysgraphia Learning Disabilities in primary school students. Since this implemented application is based on native language (Sinhala) and Android platform, in the future, we have planned to expand it to multiple languages, different mobile platform like iOS as well as a solution which will be web-based also. As planned future work, we will hope to provide a better solution which fulfils these requirements.

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